

Cardiovascular Risk Reduction

Dr. Muhammad Reihan, Internal Medicine | Source: Reihan_CVD_RR.pdf (39 slides)

Legend: ★ = high-yield / likely exam target · [added] = not on slides. Everything else comes from the lecture.

Slides 21 ("If You Quit Smoking Right Now") and 24 (HTN questions) have **no answers on the slide**: slide 21's signposts are blank and slide 24's questions are unanswered, so the content was probably delivered verbally. Those gaps are flagged below.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Classify modifiable vs non-modifiable ASCVD risk factors	"Which is non-modifiable?"
Knowledge	★ 10-year ASCVD risk categories and how they guide therapy	Cut-offs 5 / 7.5 / 20%
Skill	Calculate 10-year ASCVD risk → statin & aspirin decisions	Case vignette (as in the lecture's closing case)
Skill	Individualized plan: lifestyle + pharmacotherapy	Statin eligibility, aspirin USPSTF grades

Lecture flow: Overview/epidemiology → risk factors → 10-y ASCVD categories → Lifestyle (diet → physical activity → tobacco) → HTN → Hyperlipidemia → DM → Aspirin → Summary → Case.

2. Main Content (Lecture Order)

2.1 Overview & Epidemiology (Slides 5-8)

Critical-thinking prompt: how many relatives have had a heart attack or stroke, and how did it affect their lives?

- **CHD → > 2.5 million hospitalizations/year** worldwide
- US: an **MI every 34 seconds; one death each minute** from a major coronary event
- ★ CVD = **leading cause of mortality globally**. It includes **HTN, CAD, HF, congenital defects, strokes**.

★ Why MI death rates have declined:

1. Heightened public awareness
2. ↑ Availability of **AEDs**
3. Improved CV diagnosis & therapy
4. CV drugs in high-risk persons
5. Improved revascularization
6. More aggressive risk-factor modification

🔄 **Recap: Overview.** CVD is the #1 killer (MI every 34 s in the US). Deaths are falling because of awareness, AEDs, better diagnosis, drugs, revascularization, and risk-factor control.

🔗 **Link:** The last reason on that list, risk-factor modification, is the subject of the whole lecture. To modify risk, first identify it.

▶ **Up next: Traditional ASCVD risk factors.** Which can be changed and which cannot.

2.2 Traditional ASCVD Risk Factors (Slide 9)

Non-modifiable	Modifiable
Advancing age	Smoking
Male sex	Diabetes mellitus
Race and ethnicity	Hypertension
	Dyslipidemia (↑ total cholesterol, ↓ HDL)

🔄 **Recap: Risk factors.** Three non-modifiable (age, male sex, race/ethnicity) and four modifiable (smoking, DM, HTN, dyslipidemia).

🔗 **Link:** Individual factors don't tell you how much risk a person carries. They are combined into a single 10-year risk estimate that decides treatment.

▶ **Up next: 10-year ASCVD risk categories.**

2.3 10-Year ASCVD Risk Categories (Slide 10)

Estimates the risk of **MI or stroke over 10 years**. Used for ages ★ **40-75**. Calculated with the AHA/ACC 2013 pooled cohort calculator (MDCalc).

Category	10-y risk
Low	< 5%
Borderline	5-7.4%
Intermediate	7.5-19.9%
High	≥ 20%

Mnemonic [added]: "5 - 7.5 - 20", the three cut-offs.

🔄 **Recap: 10-y ASCVD.** Ages 40-75. Low <5, borderline 5-7.4, intermediate 7.5-19.9, high ≥20%.

🔗 **Link:** Whatever the risk category, **lifestyle is the foundation for everyone**. Drugs are added on top based on risk. The lecture therefore starts treatment with lifestyle, beginning with diet.

▶ **Up next: Diet.** Specific intake targets, the Mediterranean diet, and tailored counseling.

2.4 Lifestyle: Diet (Slides 11-15)

★ **Recommended intakes (Slide 12):**

Component	Target
Saturated fat	< 10% of total energy, replaced by polyunsaturated fat
Trans fat	< 1% of total energy
Salt	< 5 g/day
Fibre	30-45 g/day (wholegrain, fruit, vegetables)
Fruit	200 g/day (2-3 servings)
Vegetables	200 g/day (2-3 servings)
Fish	≥ 2×/week, one oily

- ★ **Mediterranean diet**: observational benefit for CV risk reduction. **PREDIMED** (NEJM 2018): primary prevention with a Mediterranean diet supplemented with **extra-virgin olive oil or nuts**.
- Nutritional counseling: food pyramid (base = daily exercise, portion & weight control, then vegetables/fruit/whole grains/healthy oils; top = limit **red meat, butter, sugary drinks, sweets, salt**).
- ★ **Tailored dietary intervention** combines 3 inputs:
 - Individual factors**: genetics, sex, age, BMI, physical activity (job/leisure), cultural habits
 - CV risk factors & comorbidities**: HTN, DM, dyslipidemia, CKD
 - CV disease**: IHD, HF

Mnemonic [added]: the "**<10, <1, <5**" rule: sat fat <10%, trans fat <1%, salt <5 g.

🔄 **Recap: Diet.** <10% saturated, <1% trans, <5 g salt, 30-45 g fibre, 200 g fruit + 200 g veg, fish twice weekly (one oily). Mediterranean diet (PREDIMED). Tailor to the individual, their risk factors, and their disease.

🔗 **Link:** Diet is energy in. Physical activity is energy out, and the second lifestyle pillar.

▶ **Up next: Physical activity.** Prescription, session structure, Saudi data.

2.5 Lifestyle: Physical Activity (Slides 16-19)

- Aerobic exercise should be offered to patients with **known CAD**
- ★ **Stable angina or stable chronic HF**: **moderate-to-vigorous** aerobic training ≥ **3×/week, 30 min/session**
- Sedentary** patients: strongly encourage starting **light-intensity** programs

★ Typical exercise session:

- Warm-up: 5-10 min**
- Aerobic conditioning: up to 30 min or more** (walking, jogging, cycling, stairs)
- Cool-down: 5-7 min**

KSA physical-activity insufficiency (MOH World Health Survey):

Highest insufficiency	Lowest insufficiency
Hail 91%, Madinah 90%, Qasim 88%	Tabuk 64%, Bahah 65%

★ Take-home:

- Lifestyle improvement should **begin early** and underlie **all** preventive efforts
- Most CV events are preventable** with lifestyle modification
- ★ **"Some is better than none; more is even better"**

🔄 **Recap: Physical activity.** $\geq 3 \times / \text{week} \times 30 \text{ min}$ moderate-vigorous for stable CAD/HF. Session = warm-up 5-10, aerobic ~30, cool-down 5-7. KSA inactivity is highest in Hail (91%). Some beats none.

🔗 **Link:** Diet and exercise are the things to add. The next lifestyle change is removing something, and it is the single most powerful one: smoking.

👉 **Up next: Tobacco cessation.** Mortality benefit, time course, and pharmacotherapy safety in CAD.

2.6 Lifestyle: Tobacco Cessation (Slides 20-22)

- ★★★ **Smoking cessation = the MOST important and MOST cost-effective** lifestyle modification for preventing CVD
- In patients with prior **MI, CABG, angioplasty, or known CHD**, quitting reduces mortality: **RR 0.64 (CI 0.58-0.71)**, i.e., ★ **36% ↓ mortality after MI**
- ★ **Time course:** mortality risk ↓ **50% in the first couple of years**. It **approaches non-smokers' risk in ~5-15 years** (Critchley & Capewell, 2012).
- Eliminating health-risk behaviours could prevent **$\geq 80\%$ of CVD** and **40% of cancers** (BACPR, 2017)
- (Slide 21 "If You Quit Smoking Right Now" has blank signposts in the PDF, so its content was not captured.)

★ Pharmacotherapy in CAD:

Agent	Safety in CAD
Nicotine replacement therapy	★ Safe , should be routinely offered
Bupropion	Safe in stable CAD (some studies)
Varenicline	Safe in some studies, but a meta-analysis found a small, statistically significant ↑ CVD

🔄 **Recap: Smoking.** Most important and most cost-effective. RR 0.64 = 36% ↓ mortality post-MI. Risk halves in ~2 years and approaches a non-smoker's in 5-15 years. NRT is safe. Varenicline has a CVD signal.

🔗 **Link:** That completes lifestyle. The lecture now turns to the modifiable disease risk factors that need targets and often drugs, starting with the #1 CV risk factor worldwide.

👉 **Up next: Hypertension.**

2.7 Hypertension (Slides 23-24)

- ★★★ **HTN = the number one cardiovascular risk factor** and the **world's greatest risk factor for death and disability**
- KSA prevalence: **mid-20% range**
- ★ People with HTN are **3× more likely to die of heart disease** and **4× more likely to die of stroke**
- ★ Target (Summary slide 35): **BP < 130/80 mm Hg for most patients**

Discussion questions posed (Slide 24), unanswered on the slide:

1. Is **home monitoring** superior to in-office monitoring?

2. What is the **optimal BP target**? (Answered on slide 35: **< 130/80**)
3. Do different **drug classes** have better outcomes?
4. Is it better to start with **one drug or more than one class**?

[added, not on slides; verify against your course: home/ambulatory BP is better than office BP for diagnosis and detects white-coat and masked HTN. BP reduction itself matters more than which first-line class (thiazide, ACEi/ARB, CCB) is used. Starting with 2 drugs is reasonable when BP is well above target (e.g., stage 2).]

🔄 **Recap: HTN.** #1 CV risk factor. Mid-20% prevalence in KSA. 3× cardiac death, 4× stroke death. Target <130/80.

🔗 **Link:** HTN damages the vessel wall. Lipids supply the material that builds the plaque, making hyperlipidemia the second major modifiable driver of atherosclerosis.

👉 **Up next: Hyperlipidemia.** Diagnosis cut-offs, risk factors, treatment, and exactly who gets a statin.

2.8 Hyperlipidemia (Slides 25-30)

- ★ One of the **most important (and modifiable)** CAD risk factors → **accelerated atherosclerosis**
- KSA: **43%** of survey respondents have **raised serum cholesterol**

Risk factors (Slide 26): diet, age, DM, inactive lifestyle, family history, gender, medications, genetic mutations

★ **Diagnosis (Slide 27):**

- Fasting or non-fasting** lipid profile (TC, HDL, TG)
- ★ Pathological values on **2 separate occasions** are needed to confirm
- ★ **Dyslipidemia = LDL > 130 mg/dL and/or HDL < 40 mg/dL**

Parameter	Optimal	Pathological
Total cholesterol	200 mg/dL (i.e., < 200)	Borderline 200-239 ; High > 240
TG	150 mg/dL (i.e., < 150)	Borderline 150-199 ; High > 200 ; Very high ≥ 500
LDL	100 mg/dL (i.e., < 100)	Near-optimal 100-129 ; Borderline high > 130 ; High > 160 ; Very high ≥ 190
HDL	≥ 60 mg/dL	Low < 40
LDL/HDL ratio	♂ < 3.5 ; ♀ < 3.0	♂ > 3.5 ; ♀ > 3.0

Management (Slide 28):

- Non-pharmacological: dietary changes, ↓ **saturated fat & cholesterol**, **eliminate alcohol**, healthy weight, physical activity
- Medical: ★ **Statins** (first line) → **PCSK9 inhibitors** (second line if poor statin response)

★★★★ **Who gets a statin? (Slide 29).** The **4 statin benefit groups** [the term "benefit groups" is added]:

#	Group
1	Clinical ASCVD

#	Group
2	LDL $\geq$ 190 mg/dL, age $\geq$ 21
3	Diabetics, LDL 70-189, age 40-75
4	10-y ASCVD risk $\geq$ 20%
(+)	ASCVD 7.5-20% + risk enhancers: family history of premature ASCVD · metabolic syndrome · CKD · inflammatory conditions · elevated biomarkers (lipoproteins, hs-CRP)

Mnemonic [added]: "**A-190-DM-20**": **ASCVD**, LDL **190**, **DM** 40-75, risk **20%**.

★ Take-home (Slide 30):

- **First assess ASCVD risk**
- Statins + lifestyle in higher-risk patients; **maximally tolerated dose** for the highest risk (clinical ASCVD)
- **PCSK9i reduce events when added to statins** in **very-high-risk secondary-prevention** patients
- ★ "**Lower is better**"

🔄 **Recap: Hyperlipidemia.** Diagnose on 2 occasions (LDL >130 and/or HDL <40). Statin groups: ASCVD, LDL ≥ 190 , DM 40-75 with LDL 70-189, risk $\geq 20\%$ (or 7.5-20% + enhancers). Statin first, then PCSK9i. Lower is better.

🔗 **Link:** Diabetes appeared inside the statin criteria, and it is itself a major CV risk multiplier. The lecture now asks how treating DM itself can protect the heart.

▶ **Up next: Diabetes mellitus.** CV risk, and the cardioprotective drug classes (SGLT-2i, GLP-1 RA).

2.9 Diabetes Mellitus (Slides 31-33)

- ★ DM $\rightarrow$ **2-3 $\times$ $\uparrow$ CVD risk**
- **CVD is a more likely cause of hospitalization** in diabetics than diabetes-related causes
- ★ Experts recommend **prevention of CV events as the goal** of diabetes treatment
- ★ **SGLT-2 inhibitors** and **GLP-1 receptor agonists** showed **CV protection** in outcome trials (T2DM with high CV risk)
- ★ Consider them when **ASCVD, HF, or CKD predominates, independent of HbA1c**
- **Dapagliflozin** (SGLT-2i) is added if **metformin monotherapy** fails to reach **HbA1c $< 7\%$** . In the lecture's case (HbA1c **6.2%** on metformin, no ASCVD/HF/CKD) it is **not needed**.
- ADA: **lifestyle management + medications**, including nutrition therapy, physical activity, smoking cessation

🔄 **Recap: DM.** 2-3 $\times$ CVD risk. The goal of treatment is preventing CV events. SGLT-2i/GLP-1 RA for ASCVD, HF, or CKD regardless of HbA1c. Otherwise add them only if HbA1c is $\geq 7\%$ on metformin.

🔗 **Link:** The last preventive tool is the most commonly misused: aspirin. Unlike statins, its benefit is small and it carries a bleeding risk, so the rules are restrictive.

▶ **Up next: Aspirin for primary prevention (USPSTF 2023).**

2.10 Aspirin for Primary Prevention (Slide 34, USPSTF 2023)

Population	Recommendation	Grade
Adults 40-59 with $\geq 10\%$ 10-y CVD risk	Individual decision. Net benefit small . More likely to benefit if not at $\uparrow$ bleeding risk and willing to take it daily.	★ C
Adults ≥ 60	Recommend AGAINST initiating low-dose aspirin for primary prevention	★ D

Mnemonic [added]: "**40-59 & 10 = maybe (C); 60+ = don't (D).**"

🔄 **Recap: Aspirin.** Primary prevention: 40-59 with $\geq 10\%$ risk is an individual choice (C). Age ≥ 60 : don't start (D). Avoid in low-risk patients and the elderly.

🔗 **Link:** The lecture now pulls everything into a summary and tests it with a case that combines ASCVD risk, DM, lipids, and aspirin.

👉 **Up next: Summary + case.**

2.11 Summary (Slide 35)

- CVD is the **leading cause of death worldwide**
- Assess every patient's CV risk**
- ★ **Lifestyle = the most important component** of prevention and risk-factor control ("**Life's Simple 7**")
 - [added: AHA Life's Simple 7 = don't smoke, be active, eat healthy, lose weight, manage BP, control cholesterol, reduce blood sugar. AHA updated it to "Life's Essential 8" (adds sleep) in 2022.]
- Statins** + lifestyle in higher-risk patients; **PCSK9i** for the **very highest** risk
- ★ **BP target $< 130/80$** for most
- ★ **Low-dose aspirin only in higher-risk** patients **if benefit $>$ bleeding risk. Avoid in low-risk patients and the elderly.**

3. High-Yield Comparison Tables

Table A: Risk factors

Non-modifiable	Modifiable (traditional)	Modifiable (lifestyle)
Age, male sex, race/ethnicity	Smoking, DM, HTN, dyslipidemia	Diet, physical inactivity, alcohol, weight

Table B: 10-y ASCVD risk $\rightarrow$ action (lecture content combined)

Risk	Statin	Aspirin (primary prevention)
$< 5\%$ (low)	Lifestyle only (unless DM/LDL ≥ 190 /ASCVD)	Avoid
5-7.4% (borderline)	Lifestyle (unless another statin group applies)	Avoid
7.5-19.9% (intermediate)	Statin if risk enhancers	$\geq 10\%$ and age 40-59 $\rightarrow$ individual (C)

Risk	Statin	Aspirin (primary prevention)
≥ 20% (high)	★ Statin	≥10% and age 40–59 → individual (C)
Any risk, age ≥ 60	Per criteria	★ Against (D)

Table C: Lipid cut-offs (see §2.8)

Table D: Statin vs PCSK9i vs aspirin vs SGLT-2i/GLP-1 RA

Drug	Who (per lecture)	Key point
Statin	4 benefit groups + intermediate risk with enhancers	First line; max tolerated dose if clinical ASCVD
PCSK9i	Poor statin response / very-high-risk secondary prevention	Add-on to statin
Aspirin	40–59, ≥10% risk, low bleeding risk	C / D grades
SGLT-2i / GLP-1 RA	T2DM with ASCVD, HF, or CKD	Independent of HbA1c

Table E: Smoking-cessation timeline

Time after quitting	Effect
First couple of years	Mortality risk ↓ 50%
5–15 years	Approaches non-smoker risk
Post-MI (overall)	RR 0.64 → 36% ↓ mortality

Table F: Lifestyle prescriptions

Domain	Numbers
Diet	Sat fat <10%, trans <1%, salt <5 g, fibre 30–45 g, fruit 200 g, veg 200 g, fish ≥2×/wk
Exercise	≥3×/wk × 30 min moderate-vigorous; warm-up 5–10, aerobic ~30, cool-down 5–7
BP	<130/80

Table G: KSA statistics

Metric	Value
HTN prevalence	Mid-20% range
Raised cholesterol	43%
Physical-activity insufficiency, highest	Hail 91%, Madinah 90%, Qasim 88%
Physical-activity insufficiency, lowest	Tabuk 64%, Bahah 65%

4. Rules, Exceptions, Patterns

- **Lifestyle for everyone, first and always.** Drugs are added according to risk.
- **The 10-y ASCVD calculator is for ages 40–75.** Exceptions: **LDL ≥190 (age ≥21)** and **clinical ASCVD** get statins **without** calculating risk.
- **DM aged 40–75 with LDL 70–189 → statin**, even if 10-y risk is low.

- **Intermediate risk (7.5-20%) → statin only with risk enhancers.**
- **Dyslipidemia needs 2 abnormal measurements**, fasting or non-fasting.
- **Aspirin for primary prevention is the exception, not the rule.** Consider only at 40-59 with $\geq 10\%$ risk and low bleeding risk. **Never start at ≥ 60 .**
- **SGLT-2i/GLP-1 RA:** the indication in ASCVD, HF, or CKD is **independent of HbA1c**. Otherwise escalate only if HbA1c $\geq 7\%$ on metformin.
- **Smoking cessation is the #1 lifestyle measure; HTN is the #1 risk factor.** Don't mix them up.
- **NRT is safe in CAD.** Varenicline has a small CVD signal (meta-analysis).
- **Lipids: "lower is better."**

5. Clinical Correlations

- **Post-MI smoker:** quitting gives a 36% mortality reduction, larger than many drugs [*comparison added*]. Offer NRT routinely.
- **Diabetic with established ASCVD/HF/CKD:** add SGLT-2i/GLP-1 RA even if HbA1c is at target.
- **Patient with LDL ≥ 190** (e.g., familial hypercholesterolemia [*added*]): high-intensity statin from age 21, no calculator needed.
- **Elderly patient asking to start aspirin "for the heart":** USPSTF D, do not start (bleeding outweighs benefit). [*added: this concerns primary prevention only. Aspirin in established CAD (secondary prevention) remains indicated, per the CAD lecture.*]
- **Stable CAD/HF patient:** prescribe structured aerobic exercise ($\geq 3 \times /wk$, 30 min) as cardiac rehabilitation.
- **KSA context:** high inactivity ($\geq 88\%$ in Hail/Madinah/Qasim) and 43% raised cholesterol make population-level lifestyle measures the priority.

6. Rapid Revision (Last-Day)

- CVD is the #1 global killer. US: an MI every **34 s**. CHD: **>2.5 M** hospitalizations/year.
- Non-modifiable: **age, male sex, race**. Modifiable: **smoking, DM, HTN, dyslipidemia**.
- 10-y ASCVD (age **40-75**): **<5 low · 5-7.4 borderline · 7.5-19.9 intermediate · ≥ 20 high**.
- Diet: **sat <10%, trans <1%, salt <5 g, fibre 30-45 g, fruit 200 g, veg 200 g, fish $\geq 2 \times /wk$ (1 oily)**. Mediterranean diet (**PREDIMED**).
- Exercise: **$\geq 3 \times /wk \times 30$ min**. Warm-up **5-10**, cool-down **5-7**. "Some > none; more > some."
- Smoking cessation = **most important and cost-effective. RR 0.64 (36% ↓)**. Risk halves in ~2 years and equals a non-smoker's in **5-15 y**. **NRT safe**.
- Eliminating risk behaviours prevents **$\geq 80\%$ of CVD** and **40% of cancers**.
- HTN = **#1 CV risk factor**. **3x** heart death, **4x** stroke death. Target **<130/80**. KSA **mid-20%**.
- Lipids: dyslipidemia = **LDL >130 and/or HDL <40**, confirmed on **2 occasions**. KSA **43%** raised cholesterol.
- Statin groups: **ASCVD · LDL ≥ 190 (≥ 21 y) · DM 40-75 + LDL 70-189 · risk $\geq 20\%$ · 7.5-20% + enhancers**.
- PCSK9i = add-on for very-high-risk patients. **Lower is better**.
- DM = **2-3x** CVD. **SGLT-2i/GLP-1 RA** if ASCVD/HF/CKD, **regardless of HbA1c**.
- Aspirin (USPSTF 2023): **40-59 & $\geq 10\%$ → C (individual); ≥ 60 → D (against)**.
- **Life's Simple 7.**

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slides 36-37). 54 ♂, T2DM on metformin, **HbA1c 6.2%**, diverticulosis, maternal aunt had an MI at 63, non-smoker, 3-4 beers/week, BMI 24, **BP 124/68**. TC 185, **HDL 45, LDL 122**, TG 140. **10-y risk 9.1%**. Besides lifestyle counseling, the most appropriate recommendation? A) Aspirin **B) Atorvastatin** C) Dapagliflozin D) No further action E) PCSK9i

Answer: B. He is **diabetic, age 40-75, LDL 70-189 (122)** → statin benefit group regardless of risk %.

- **A X:** 10-y risk **9.1% < 10%**, so he doesn't meet the USPSTF threshold.
- **C X:** HbA1c **6.2% < 7%** on metformin with **no ASCVD, HF, or CKD**.
- **D X:** he meets a statin indication.
- **E X:** PCSK9i only for poor statin response or very-high-risk secondary prevention.
- Distractor: the **aunt is not a first-degree relative** (and 63 is not "premature" for a female), so this is not a family-history risk enhancer *[reasoning added]*.

Q2 (MCQ). Which lifestyle intervention is described as the **most important and most cost-effective** for preventing CVD? a) Mediterranean diet b) Aerobic exercise $\geq 3\times/\text{week}$ **c) Smoking cessation** d) Salt < 5 g/day **Answer: c.** Stated explicitly (Slide 20). Don't confuse it with "HTN = #1 risk factor".

Q3 (MCQ). A 65-year-old woman with a 10-y ASCVD risk of 12% and no bleeding history asks whether to start low-dose aspirin for primary prevention. Best advice? **Answer: Recommend against (USPSTF D, age ≥ 60).** The $\geq 10\%$ risk criterion applies only to ages **40-59** (grade C).

Q4 (Short answer). List the statin indications taught in the lecture. **Answer:** (1) Clinical ASCVD. (2) LDL ≥ 190 mg/dL, age ≥ 21 . (3) Diabetics aged 40-75 with LDL 70-189. (4) 10-y ASCVD risk $\geq 20\%$. (5) Risk 7.5-20% plus risk enhancers: family history of premature ASCVD, metabolic syndrome, CKD, inflammatory conditions, elevated biomarkers (lipoproteins, hs-CRP).

Q5 (MCQ). A patient's lipid profile shows LDL 145 mg/dL and HDL 50 mg/dL on one measurement. Next step? **Answer: Repeat the lipid profile on a second occasion** before diagnosing dyslipidemia (2 pathological values are required). LDL 145 is "borderline high" (>130).

8. Exam Pitfalls & Traps

1. **"#1 risk factor" (HTN) vs "most important/cost-effective intervention" (smoking cessation).** These are different questions.
2. **Aspirin for everyone at risk.** No: 40-59 with $\geq 10\%$ risk only (C). **≥ 60 is D.** Low-risk patients: avoid.
3. **Applying the 10-y calculator outside ages 40-75,** or to patients with LDL ≥ 190 or ASCVD (they get statins regardless).
4. **Forgetting the DM statin rule.** A diabetic aged 40-75 with LDL 70-189 needs a statin even if the 10-y risk is under 7.5%.
5. **Intermediate risk $\neq$ automatic statin.** Risk enhancers are required.
6. **Adding an SGLT-2i because the patient "has diabetes."** It is needed only with ASCVD/HF/CKD or HbA1c $\geq 7\%$ on metformin.

7. **Diagnosing dyslipidemia from a single value.** Two occasions are required. Fasting is **not** mandatory.
8. **Premature family history** means a **first-degree** relative (σ <55, ♀ <65, per the CAD lecture). Aunts, uncles and grandparents don't count.
9. **Varenicline "totally safe" in CAD.** The meta-analysis showed a small $\uparrow$ CVD. **NRT is the safe routine choice.**
10. **Confusing risk bands: 7.5%** (statin threshold with enhancers) vs **10%** (aspirin threshold) vs **20%** (high risk, statin).
11. **Time course of smoking benefit:** 50% $\downarrow$ within $\sim$ 2 years, but **5-15 years** to reach non-smoker levels. It is not immediate.